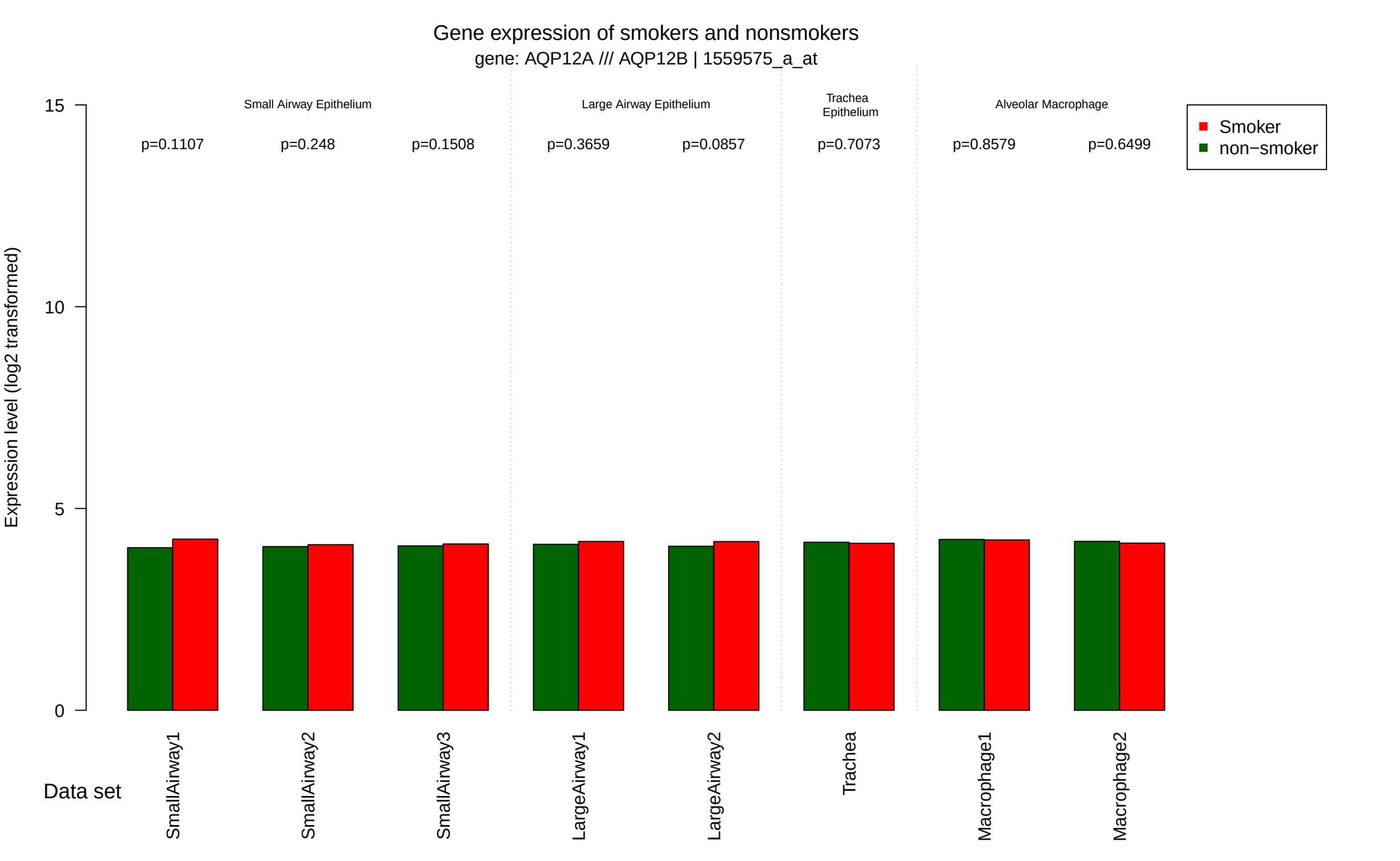
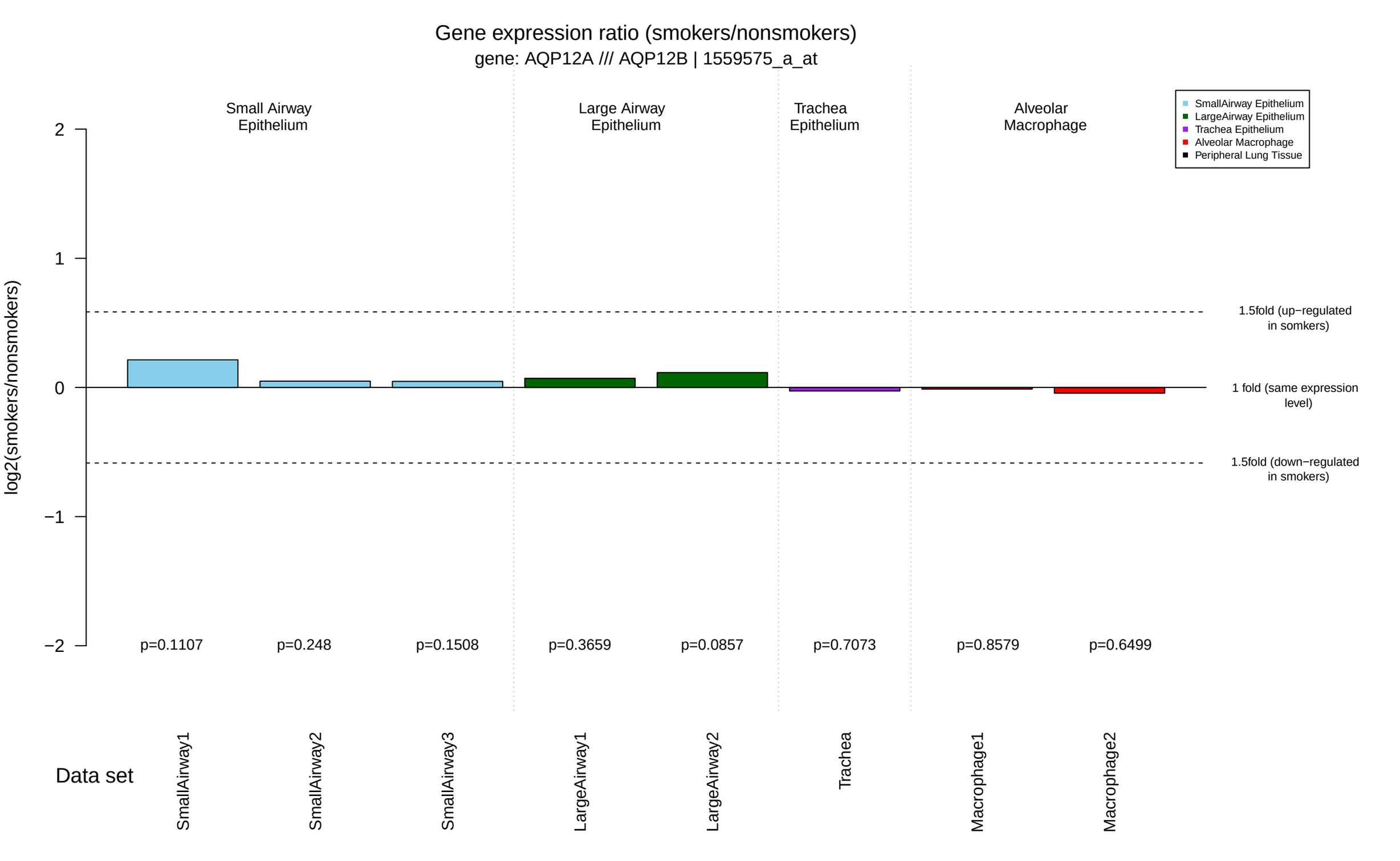
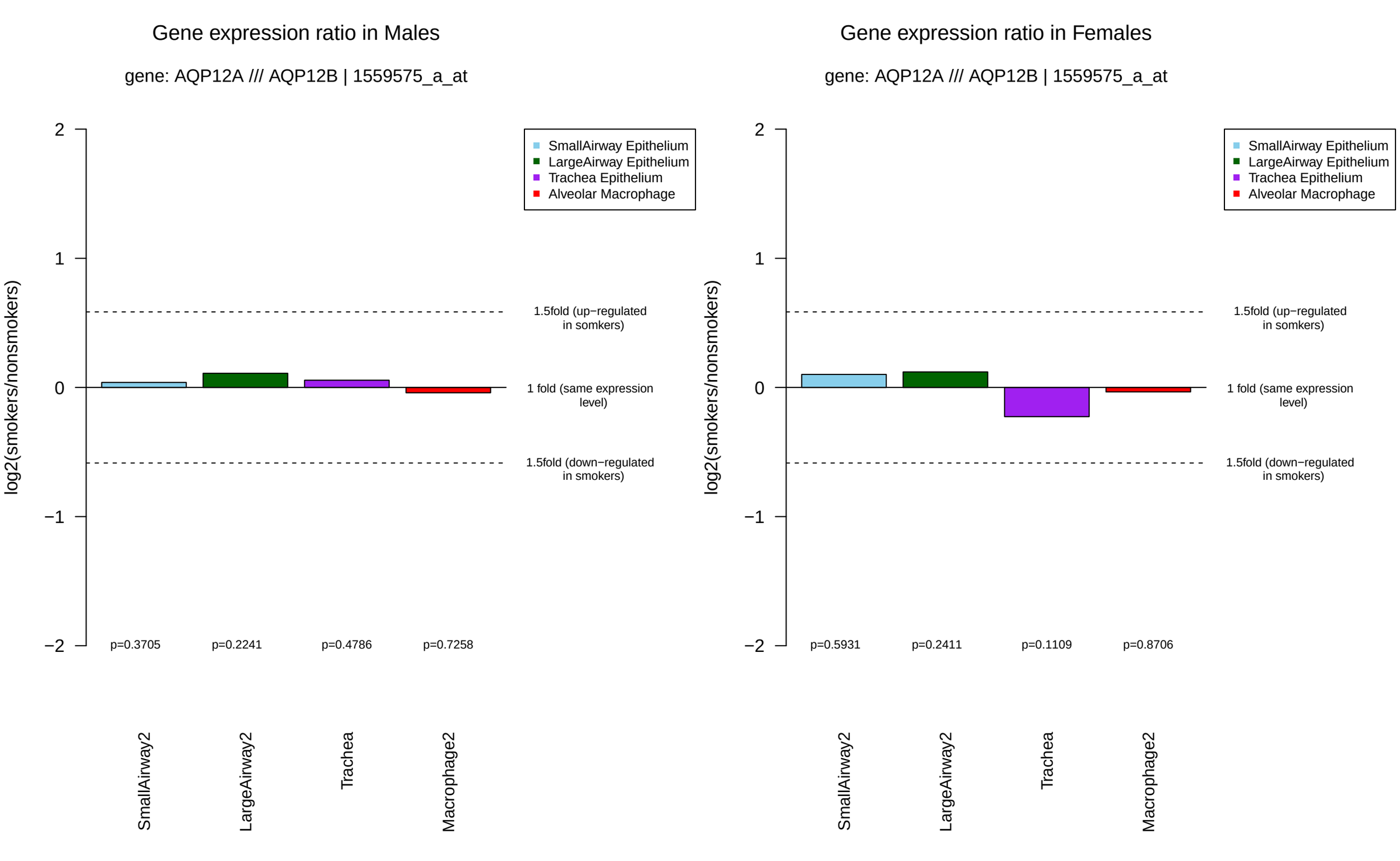
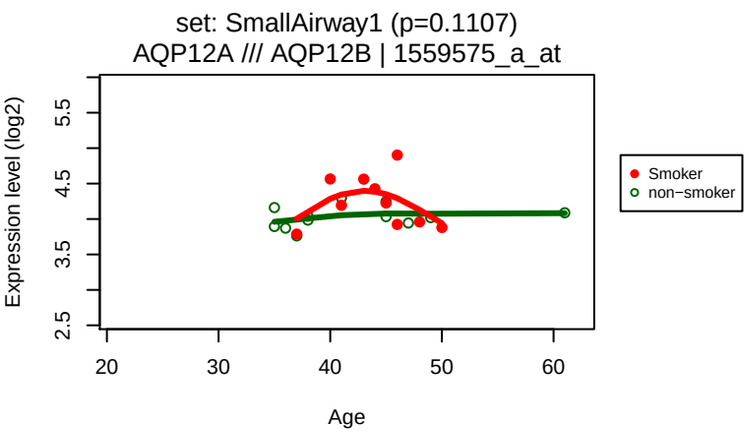
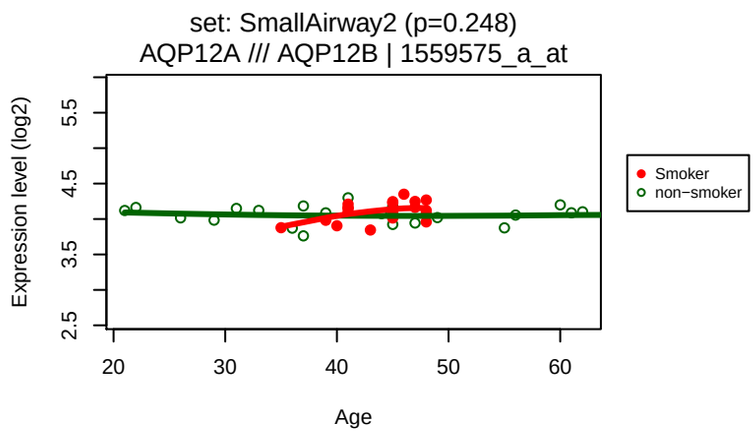
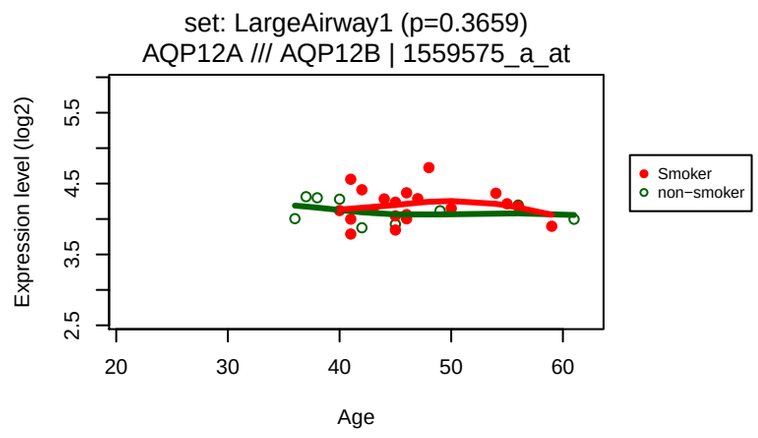

Gene expression of smokers and nonsmokers
gene: AQP12A /// AQP12B | 1559575_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1107
p=0.248
p=0.1508
p=0.3659
p=0.0857
p=0.7073
p=0.8579
p=0.6499
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

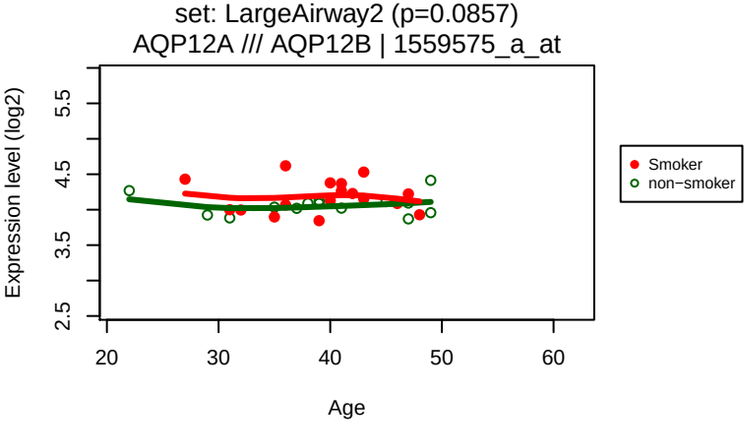

Gene expression ratio (smokers/nonsmokers)
gene: AQP12A /// AQP12B | 1559575_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1107
p=0.248
p=0.1508
p=0.3659
p=0.0857
p=0.7073
p=0.8579
p=0.6499
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

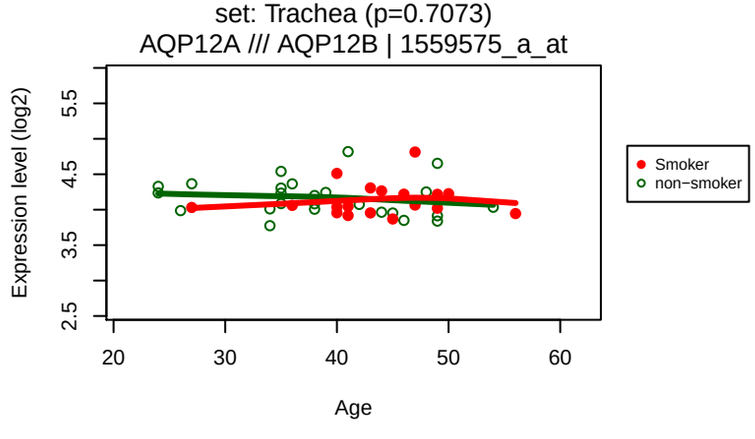

Gene expression ratio in Males
gene: AQP12A /// AQP12B | 1559575_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3705
p=0.2241
p=0.4786
p=0.7258
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP12A /// AQP12B | 1559575_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5931
p=0.2411
p=0.1109
p=0.8706
SmallAirway2
LargeAirway2
Trachea
Macrophage2

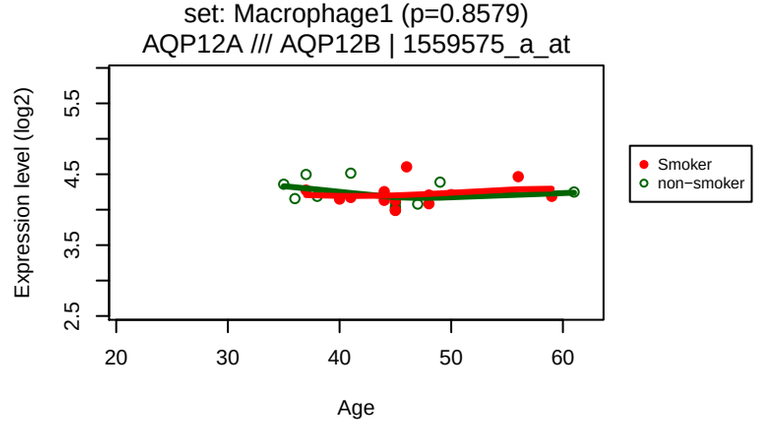

set: SmallAirway1 (p=0.1107)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

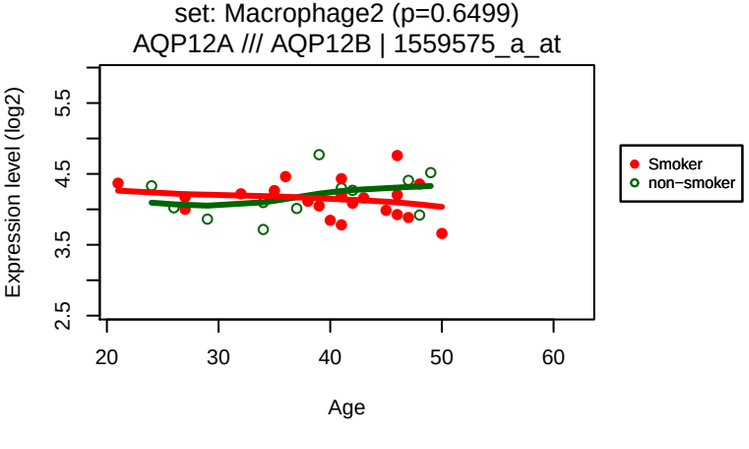

set: SmallAirway2 (p=0.248)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

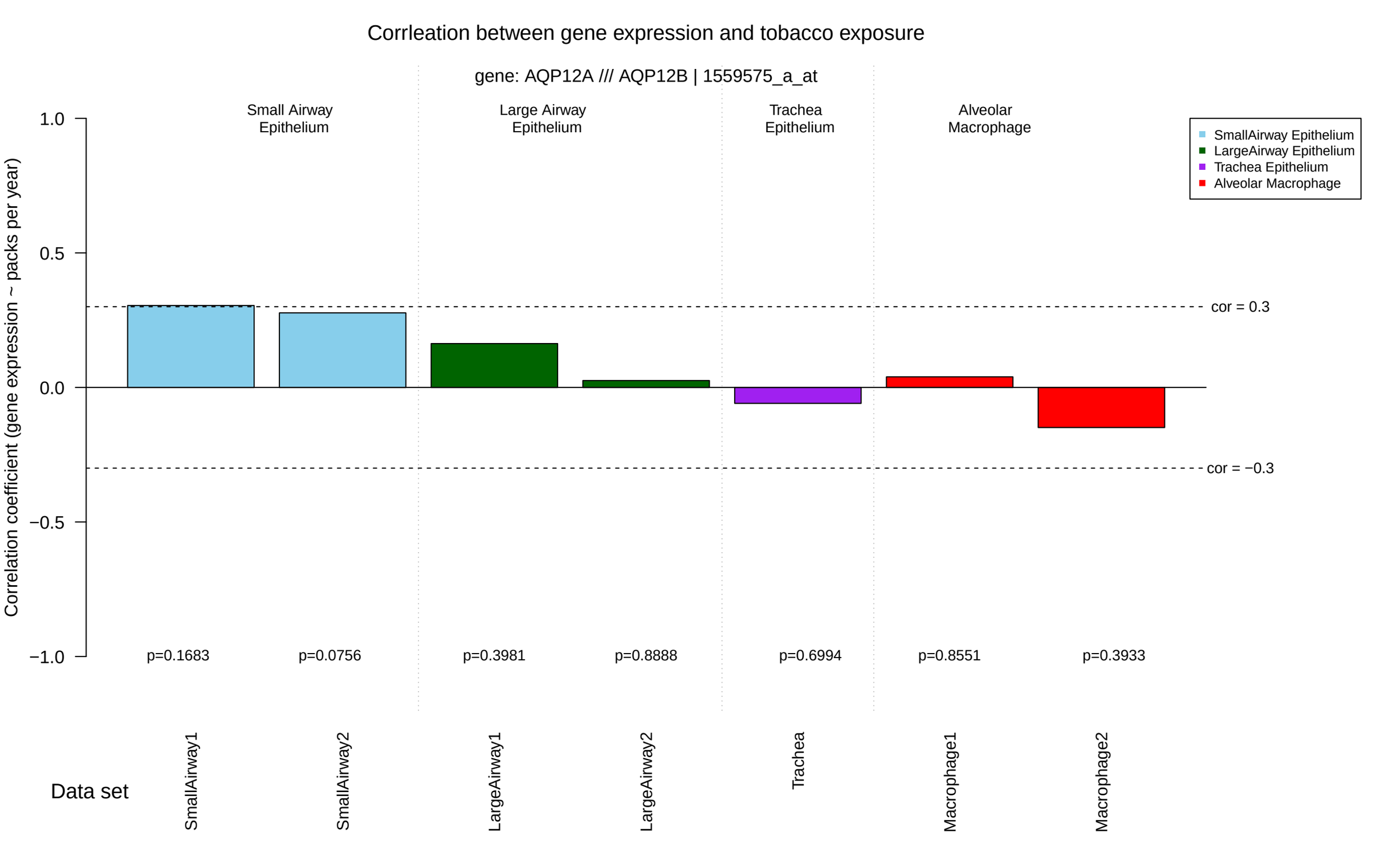

set: LargeAirway1 (p=0.3659)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0857)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7073)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8579)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6499)
AQP12A /// AQP12B | 1559575_a_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP12A /// AQP12B | 1559575_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1683
p=0.0756
p=0.3981
p=0.8888
p=0.6994
p=0.8551
p=0.3933
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2